Khalil Zaghla

Final-Year Engineering Student | Computer Vision & Applied AI
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SUMMARY

Final-year Data Engineering student at ENET'Com specializing in Computer Vision, Deep Learning, and real-time video intelligence. Hands-on experience developing spatio-temporal action recognition pipelines, pose estimation models, and distributed ML architectures (PyTorch, YOLO, SlowFast, VideoMAE). Seeking a Computer Vision graduation internship (PFE) / junior role to build and deploy robust, production-oriented vision systems.

PROFESSIONAL EXPERIENCE

Computer Vision Intern — Retail Loss Prevention 2026
Anavid

- Designed a privacy-preserving shoplifting detection pipeline using YOLO-Pose, ByteTrack, and robust temporal skeleton representations based on relative coordinates, torso normalization, interpolation, and smoothing.
- Developed and evaluated STG-NF, ST-LSTM, and Transformer models for behavioral anomaly detection, while benchmarking pose-estimation latency–performance trade-offs and tracking experiments with Weights & Biases.

YOLO-Pose · ByteTrack · STG-NF · ST-GCN · ST-LSTM · Transformers · PyTorch · W&B

Computer Vision Intern — Crab Species Detection 2025
Digital Research Center of Sfax

- Engineered an end-to-end two-stage vision pipeline using YOLOv8 detection and transfer-learned CNNs (ResNet, EfficientNet) for species and sex classification on custom web-scraped, annotated, and augmented datasets.

YOLOv8 · ResNet · EfficientNet · Transfer Learning · Web Scraping · PyTorch · Roboflow · Python

PROJECTS

Spatio-Temporal Action Recognition & Video Tracking Pipeline 2026

- Engineered an end-to-end surveillance pipeline combining YOLOv8 tracking with SlowFast (ResNet-50 3D) and 3D RoIAlign for actor-centric anomaly classification, deploying an asynchronous Flask multi-stream dashboard.

PyTorch · YOLOv8 · SlowFast (3D CNN) · 3D RoIAlign · OpenCV · Flask · Docker

Reinforcement Learning for Video Anomaly Scoring (VideoMAE + PPO) 2026

- Developed an active anomaly scoring framework applying PPO (Actor-Critic) over fine-tuned VideoMAE embeddings, with custom rewards optimizing early detection and localization without frame-level labels.

PyTorch · VideoMAE · Reinforcement Learning · PPO · Transformers · Hugging Face

Real-Time Streaming Analytics & Graph Fraud Detection Platform 2025

- Built a streaming pipeline processing 50k+ events/hour with sub-200 ms latency into ClickHouse, serving a GraphSAGE GNN model achieving 92% F1-score for real-time transaction inference.

Kafka · Spark · ClickHouse · GNN (GraphSAGE) · Airflow · Docker · Streamlit

SKILLS

Computer Vision: YOLO (v8, Pose), SlowFast, 3D RoIAlign, VideoMAE, ByteTrack, ST-GCN, STG-NF, OpenCV, 3D CNNs, Pose Estimation
Deep Learning: PyTorch, Reinforcement Learning (PPO, Actor-Critic), Transformers, GNNs, Transfer Learning, W&B
Data Engineering: Kafka, Spark, ClickHouse, Airflow, SQL, PostgreSQL, ETL/ELT, Streamlit
Deployment & Systems: Docker, FastAPI, Flask, Multithreading, Git, Linux

EDUCATION

National School of Electronics and Telecommunications of Sfax (ENET'Com) 2024 – Present
Engineering Degree in Data Engineering and Decision Systems

Preparatory Institute for Engineering Studies of Sfax (IPEIS) 2021 – 2023
Physics and Chemistry Preparatory Program

LEADERSHIP & ACTIVITIES

Google Developer Student Club (GDSC) – ENET'Com 2025
Machine Learning Lead

- Led 13+ technical sessions on ML, deep learning, and generative AI for 40+ students, with hands-on labs covering training, evaluation, and deployment.
- Contributed to the technical challenge for the MindShift 2.0 AI Hackathon, defining the ML problem, preparing datasets, and supporting participants.

LANGUAGES

Arabic: Native | English: Professional | French: Professional